

Task 3: Estimation and interpretation of SAOMs

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Task 3.1.1

We compute the Jaccard index to evaluate whether the data contain enough information to investigate the evolution of the friendship network.

First we load the three friendship networks

```
f1 <- as.matrix(read.csv("Glasgow/f1.csv", header = FALSE))
f2 <- as.matrix(read.csv("Glasgow/f2.csv", header = FALSE))
f3 <- as.matrix(read.csv("Glasgow/f3.csv", header = FALSE))
```

We can calculate the Jaccard index as follows: $J(t_i, t_i + 1) = \frac{N_{11}}{N_{01} + N_{10} + N_{11}}$, where t_i is the wave, and N_{11} is the number of ties that are present in both waves, N_{10} and N_{01} are number of ties present only at wave i and $i + 1$ respectively.

Now we create a function which takes the friendship networks and outputs their Jaccard index:

```
jaccard <- function(A, B) {
  N11 <- sum(A == 1 & B == 1)
  N10 <- sum(A == 1 & B == 0)
  N01 <- sum(A == 0 & B == 1)
  N11 / (N11 + N10 + N01)
}

J12 <- jaccard(f1, f2)
J23 <- jaccard(f2, f3)
```

The Jaccard index for wave 1 -> wave 2 is

```
## [1] 0.3036496
```

and for wave 2 -> wave 3 is

```
## [1] 0.3506686
```

We can also calculate the number of ties that changed during two consecutive networks.

For the transition from wave 1 to wave 2 we have:

```
sum(f1 == 0 & f2 == 1) + sum(f1 == 1 & f2 == 0)
```

```
## [1] 477
```

And for the transition from wave 2 to wave 3 we have:

```
sum(f2 == 0 & f3 == 1) + sum(f2 == 1 & f3 == 0)
```

```
## [1] 437
```

Comments: Both wave transitions have Jaccard indices above the threshold of 0.3, (0.304 from wave 1 to wave 2 and 0.351 from wave 2 to wave 3) which indicates moderate similarity between the waves. If the Jaccard index was too low, then the two waves would be too different to have a well-identified continuous-time Markov process assumed by the SAOM. If the Jaccard index was too high, then that would mean too few changes to estimate the parameters from. The indices indicate moderate stability in the friendship network across waves, with enough tie changes to estimate the model parameters reliably. The 477 and 437 tie changes across the two intervals that we calculated provide enough variation to estimate the model. We observe a slight increase from 0.304 to 0.351, which shows that the transition from wave 2 to wave 3 is more stable than the transition from wave 1 to wave 2, however this difference is quite small. Overall, the data contain sufficient information to investigate the evolution of the friendship network with a SAOM.

Task 3.1.2

We want to test the following three hypotheses:

- i. Students tend to be friends with popular pupils
- ii. Students tend to be friends with pupils with similar alcohol consumption to theirs
- iii. Students tend to be friends with students who live in the same neighborhood (living nearby)

```
library(RSiena)
library(sna)
```

```
## Loading required package: statnet.common
```

```
##
```

```
## Attaching package: 'statnet.common'
```

```

## The following objects are masked from 'package:base':
##
## attr, order, replace

## Loading required package: network

##
## 'network' 1.20.0 (2026-02-06), part of the Statnet Project
## * 'news(package="network")' for changes since last version
## * 'citation("network")' for citation information
## * 'https://statnet.org' for help, support, and other information

## sna: Tools for Social Network Analysis
## Version 2.8 created on 2024-09-07.
## copyright (c) 2005, Carter T. Butts, University of California-Irvine
## For citation information, type citation("sna").
## Type help(package="sna") to get started.

library(parallel)

source("printSiena.R")

# Load data
f1 <- as.matrix(read.csv("Glasgow/f1.csv", header = FALSE))
f2 <- as.matrix(read.csv("Glasgow/f2.csv", header = FALSE))
f3 <- as.matrix(read.csv("Glasgow/f3.csv", header = FALSE))

demographic <- read.csv("Glasgow/demographic.csv", header = TRUE)
alcohol <- as.matrix(read.csv("Glasgow/alcohol.csv", header = TRUE))
logdistance <- as.matrix(read.csv("Glasgow/logdistance.csv", header = FALSE))

# Create Siena dependent network object
friendship <- sienaDependent(array(c(f1, f2, f3), dim = c(129, 129, 3)))

# Covariates
gender <- coCovar(demographic$gender)
age <- coCovar(demographic$age)
alcoholCov <- varCovar(alcohol)
distance <- coDyadCovar(logdistance)

# Create Siena data object
mydata <- sienaDataCreate(friendship, gender, age, alcoholCov, distance)

# Start model specification
myeff <- getEffects(mydata)

# Basic endogenous controls
myeff <- includeEffects(myeff, recip, transTrip, cycle3, name = "friendship")

## effectNumber effectName shortName include fix
## 1 18 reciprocity recip TRUE FALSE
## 2 24 transitive triplets transTrip TRUE FALSE
## 3 44 3-cycles cycle3 TRUE FALSE

```

```
## test initialValue parm
## 1 FALSE      0 0
## 2 FALSE      0 0
## 3 FALSE      0 0
```

```
# Basic exogenous controls: gender
myeff <- includeEffects(myeff, egoX, altX, sameX,
                        interaction1 = "gender",
                        name = "friendship")
```

```
## effectNumber effectName shortName include fix test
## 1 298         gender alter altX      TRUE  FALSE FALSE
## 2 313         gender ego   egoX      TRUE  FALSE FALSE
## 3 369         same gender sameX     TRUE  FALSE FALSE
## initialValue parm
## 1      0 0
## 2      0 0
## 3      0 0
```

```
# Basic exogenous controls: age
myeff <- includeEffects(myeff, egoX, altX, simX,
                        interaction1 = "age",
                        name = "friendship")
```

```
## effectNumber effectName shortName include fix test
## 1 482         age alter altX      TRUE  FALSE FALSE
## 2 497         age ego   egoX      TRUE  FALSE FALSE
## 3 543         age similarity simX   TRUE  FALSE FALSE
## initialValue parm
## 1      0 0
## 2      0 0
## 3      0 0
```

```
# Hypothesis i:
myeff <- includeEffects(myeff, inPop,
                        name = "friendship")
```

```
## effectNumber effectName shortName include
## 1 81          indegree - popularity inPop TRUE
## fix test initialValue parm
## 1 FALSE FALSE      0 0
```

```
# Hypothesis ii:
myeff <- includeEffects(myeff, egoX, altX, simX,
                        interaction1 = "alcoholCov",
                        name = "friendship")
```

```
## effectNumber effectName shortName include
## 1 666         alcoholCov alter altX      TRUE
## 2 681         alcoholCov ego   egoX      TRUE
## 3 727         alcoholCov similarity simX   TRUE
```

```
## fix test initialValue parm
## 1 FALSE FALSE 0 0
## 2 FALSE FALSE 0 0
## 3 FALSE FALSE 0 0
```

```
# Hypothesis iii:
myeff <- includeEffects(myeff, X,
                        interaction1 = "distance",
                        name = "friendship")
```

```
## effectNumber effectName shortName include fix test
## 1 265 distance X TRUE FALSE FALSE
## initialValue parm
## 1 0 0
```

```
myeff
```

```
## effectName include fix test
## 1 constant friendship rate (period 1) TRUE FALSE FALSE
## 2 constant friendship rate (period 2) TRUE FALSE FALSE
## 3 outdegree (density) TRUE FALSE FALSE
## 4 reciprocity TRUE FALSE FALSE
## 5 transitive triplets TRUE FALSE FALSE
## 6 3-cycles TRUE FALSE FALSE
## 7 indegree - popularity TRUE FALSE FALSE
## 8 distance TRUE FALSE FALSE
## 9 gender alter TRUE FALSE FALSE
## 10 gender ego TRUE FALSE FALSE
## 11 same gender TRUE FALSE FALSE
## 12 age alter TRUE FALSE FALSE
## 13 age ego TRUE FALSE FALSE
## 14 age similarity TRUE FALSE FALSE
## 15 alcoholCov alter TRUE FALSE FALSE
## 16 alcoholCov ego TRUE FALSE FALSE
## 17 alcoholCov similarity TRUE FALSE FALSE
## initialValue parm
## 1 7.45424 0
## 2 6.82927 0
## 3 -1.61299 0
## 4 0.00000 0
## 5 0.00000 0
## 6 0.00000 0
## 7 0.00000 0
## 8 0.00000 0
## 9 0.00000 0
## 10 0.00000 0
## 11 0.00000 0
## 12 0.00000 0
## 13 0.00000 0
## 14 0.00000 0
## 15 0.00000 0
## 16 0.00000 0
## 17 0.00000 0
```

Here, we specify a SAOM where the friendship network is the dependent variable and evolves over time. For the endogenous variables in the network, we include effects for reciprocity, transitive triplets and three-cycles. Three-cycles are included alongside transitive triplets to separate hierarchical transitivity from cyclic closure patterns. We also include the exogenous covariates such as gender and age. To test the hypotheses, for hypothesis i we include the indegree popularity effect to see whether students tend to form ties with more popular students. For hypothesis ii, we include the variable of alcohol similarity to test whether students tend to form ties with others who have similar alcohol consumption levels. The ego and alter effects for alcohol are included as controls to account for the possibility that students with higher or lower alcohol consumption are systematically more or less socially active or popular independently of similarity. For hypothesis iii, we use the covariate for log-distance to test whether the geographical proximity increases the likelihood of students forming a tie.

Task 3.1.3

We estimate the model:

```
# Estimate the model
myAlgorithm <- set_algorithm_saom(seed = 1908, n3 = 3000, nsub = 4)

modelEv <- siena(data = mydata, effects = myeff, returnDeps = TRUE,
                 control_algo = myAlgorithm)
```

siena will create/use an output file mydata_out.txt .

```
# Results
summary(modelEv)
```

```
## Estimates, standard errors and convergence t-ratios
##
##
##
##
## Rate parameters:
## 0.1      Rate parameter period 1 11.0285 ( 1.0278 )
## 0.2      Rate parameter period 2  9.0751 ( 0.8272 )
##
## Other parameters:
## 1. eval outdegree (density)    -3.0771 ( 0.1286 )    0.0829
## 2. eval reciprocity            2.2154 ( 0.1038 )    0.0802
## 3. eval transitive triplets     0.6900 ( 0.0441 )    0.0788
## 4. eval 3-cycles               -0.4677 ( 0.0858 )    0.0708
## 5. eval indegree - popularity  -0.0811 ( 0.0194 )    0.0707
## 6. eval distance               -0.1573 ( 0.0454 )    0.0032
## 7. eval gender alter           -0.1682 ( 0.1041 )   -0.0206
## 8. eval gender ego             0.1351 ( 0.1064 )   -0.0272
## 9. eval same gender            0.9000 ( 0.1004 )    0.0886
## 10. eval age alter             0.1171 ( 0.1166 )   -0.0131
## 11. eval age ego              -0.2146 ( 0.1130 )    0.0735
## 12. eval age similarity         0.0391 ( 0.1855 )    0.0116
## 13. eval alcoholCov alter       0.0299 ( 0.0319 )   -0.0433
## 14. eval alcoholCov ego        -0.0221 ( 0.0308 )   -0.0311
## 15. eval alcoholCov similarity  0.6376 ( 0.1628 )    0.0150
```

```

##
## Overall maximum convergence ratio:    0.1646
##
##
## Total of 4999 iteration steps.
##
## Covariance matrix of estimates (correlations below diagonal)
##
##      0.017      -0.003      0.000      0.001      -0.001      0.000      0.001      -0.001
##     -0.207      0.011      0.002     -0.005      0.000      0.000      0.000      0.000
##     -0.013      0.338      0.002     -0.003      0.000      0.000      0.000      0.000
##      0.083     -0.586     -0.775      0.007      0.000      0.000     -0.001      0.000
##     -0.578      0.035     -0.300      0.045      0.000      0.000      0.000      0.000
##     -0.044      0.043     -0.002      0.013      0.104      0.002      0.000      0.000
##      0.042     -0.001      0.025     -0.091      0.150     -0.049      0.011     -0.001
##     -0.095     -0.019     -0.054      0.104     -0.070      0.016     -0.736      0.000
##     -0.707     -0.033     -0.031      0.040     -0.003      0.011     -0.150      0.000
##     -0.024      0.089      0.045     -0.023      0.003      0.026      0.044     -0.001
##      0.057     -0.047      0.007     -0.009      0.009     -0.043      0.029     -0.001
##     -0.006      0.042      0.021     -0.050     -0.033     -0.025      0.001     -0.001
##      0.044     -0.010     -0.027      0.072     -0.072      0.024     -0.151      0.000
##     -0.006      0.020      0.051     -0.053      0.029      0.019     -0.013     -0.001
##     -0.094     -0.042     -0.050      0.030      0.037     -0.064     -0.036      0.000
##
## Derivative matrix of expected statistics X by parameters:
##
##      1126.520      741.809      2556.184      688.129      9296.291     -149.708       3.126      15.9
##       439.632      505.664      1373.254      459.019      3808.195      -83.322       5.259       3.0
##      1923.436     1594.709      9555.816     2802.385     19282.647     -370.074      34.078     43.8
##       817.975      856.176      4348.124     1464.756      8292.160     -173.755      19.548       7.1
##      5528.958     3671.947     15430.249     4131.093     52014.958     -887.321     -72.485     20.8
##      -192.291     -157.853     -633.490     -190.641     -1895.483      517.839       5.994       3.4
##       -15.641        6.133        3.624       13.426     -381.889      -4.440     271.946     233.0
##        13.314        1.425       28.154        1.076      -60.391     -10.610     233.991     280.4
##      1033.281      691.892      2263.395      601.200      8478.976     -136.117      -5.045     -5.5
##       -7.710     -20.232     -84.205     -31.463     -110.562       1.060     -8.273     -7.1
##      -25.097     -6.215     -52.213     -11.391     -226.478       7.679       0.260     -1.3
##       12.787       9.894       55.129       15.482      127.932       0.462       3.458       4.0
##        5.179      -5.418       14.747     -14.521     -65.495     -44.209     162.358     146.8
##       16.216      36.166       47.660       28.909     -32.443     -44.144     144.708     160.0
##       61.141      48.916      144.500       43.181      488.652       1.463      -4.486     -3.5
##
## Covariance matrix of X (correlations below diagonal):
##
##      1502.580      1142.573      5248.807      1494.246      13731.716     -261.519       27.390      48.5
##        0.856      1187.078      4892.834      1549.563      10791.347     -242.959       29.967      28.0
##        0.749        0.786     32660.923      9896.141      56931.841    -1209.108      216.558     241.4
##        0.684        0.798        0.972      3174.889      16363.392     -369.478       73.074      64.0
##        0.952        0.842        0.847        0.781     138386.640    -2668.100       21.289     227.0
##       -0.293     -0.306     -0.290     -0.284       -0.311      531.264     -29.398     -28.4
##        0.038        0.047        0.064        0.069        0.003      -0.068     348.890     333.4
##        0.064        0.042        0.068        0.059        0.031      -0.063       0.911     383.7
##        0.956        0.827        0.687        0.619        0.903      -0.277       0.010       0.0
##        0.032        0.000       -0.014     -0.017        0.017      -0.019       0.005       0.0

```

```
##      -0.072      -0.031      -0.054      -0.038      -0.078      0.014      0.013      -0.0
##      0.073      0.068      0.056      0.048      0.077      0.007      0.015      0.0
##      0.062      0.064      0.050      0.043      0.050      -0.084      0.338      0.3
##      0.006      0.041      0.000      0.010      -0.007      -0.064      0.314      0.3
##      0.304      0.297      0.236      0.231      0.284      -0.025      -0.021      -0.0
```

```
# Check convergence
modelEv$tconv.max
```

```
##      [,1]
## [1,] 0.1646112
```

Comments: The SAOM was estimated using a stochastic approximation algorithm with 3000 phase 3 simulations. The overall maximum convergence ratio is 0.1646, which is below 0.25 and therefore indicates satisfactory convergence. Also, good convergence is indicated by the t-ratios being close to zero.

Now let's look at the parameters: The estimated parameters in the model indicate some important patterns in the evolution of the friendship network. Statistical significance is assessed by computing the t-ratio, defined as the estimated coefficient divided by its standard error. A t-ratio with absolute value greater than 2 is generally considered statistically significant.

The estimate for density (outdegree) is -3.08 (with a standard error of 0.129). This is a significant negative outdegree density, indicating that the networks are sparse and therefore students tend to not form many ties. The estimate for reciprocity is 2.22 (with a standard error of 0.104). This is a strong positive estimator, which indicated that students strongly tend to reciprocate friendships. The estimate for transitive triplets is 0.690 (with a standard error of 0.044). This indicates a strong triadic closure, which means that students are more likely to be tied to friends of their friends. The estimate for 3-cycles is -0.468 (with a standard error of 0.086), which indicates that the network is more hierarchical than cyclic. The gender homophily effect is strongly positive, since the estimate is 0.900 (with a standard error of 0.100), indicating that students strongly prefer same-gender friendships.

Now, looking at the three hypotheses: The indegree popularity estimate is -0.081 with a standard error of 0.019. So the indegree effect is negative and statistically significant, which suggests that students are less likely to form ties with students who are popular. This means that Hypothesis i is not supported by the data. The alcohol similarity estimate is 0.638 with a standard error of 0.163. This shows that the alcohol similarity effect is positive and significant, indicating that students tend to form friendships with others that have similar alcohol consumption levels. This supports Hypothesis ii. The distance estimate is -0.157 with a standard error of 0.045. This shows a significant negative effect, indicating that students are more likely to form friendships when they live closer to each other, supporting Hypothesis iii.

The remaining control effects for age and gender are significant, confirming that they are important co-founders to account for in the model.

Task 3.1.4

We evaluate the goodness of fit of the estimated model using the following auxiliary statistics: indegree distribution, outdegree distribution, triad census and geodesic distance.

```
# Geodesic distance function
GeodesicDistribution <- function(
  i, data, sims, period, groupName,
  varName, levls = c(1:5, Inf), cumulative = TRUE
) {
  x <- networkExtraction(i, data, sims, period, groupName, varName)
```



```

require(sna)
a <- sna::geodist(symmetrize(x))$gdist

if (cumulative) {
  gdi <- sapply(levls, function(i) sum(a <= i))
} else {
  gdi <- sapply(levls, function(i) sum(a == i))
}

names(gdi) <- as.character(levls)
gdi
}

# Use parallel computation
cl <- makeCluster(4)

# Indegree distribution
gofEvId <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", IndegreeDistribution,
  cluster = cl
)

# Outdegree distribution
gofEvOd <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", OutdegreeDistribution,
  cluster = cl
)

# Triad census
gofEvTC <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", TriadCensus,
  cluster = cl
)

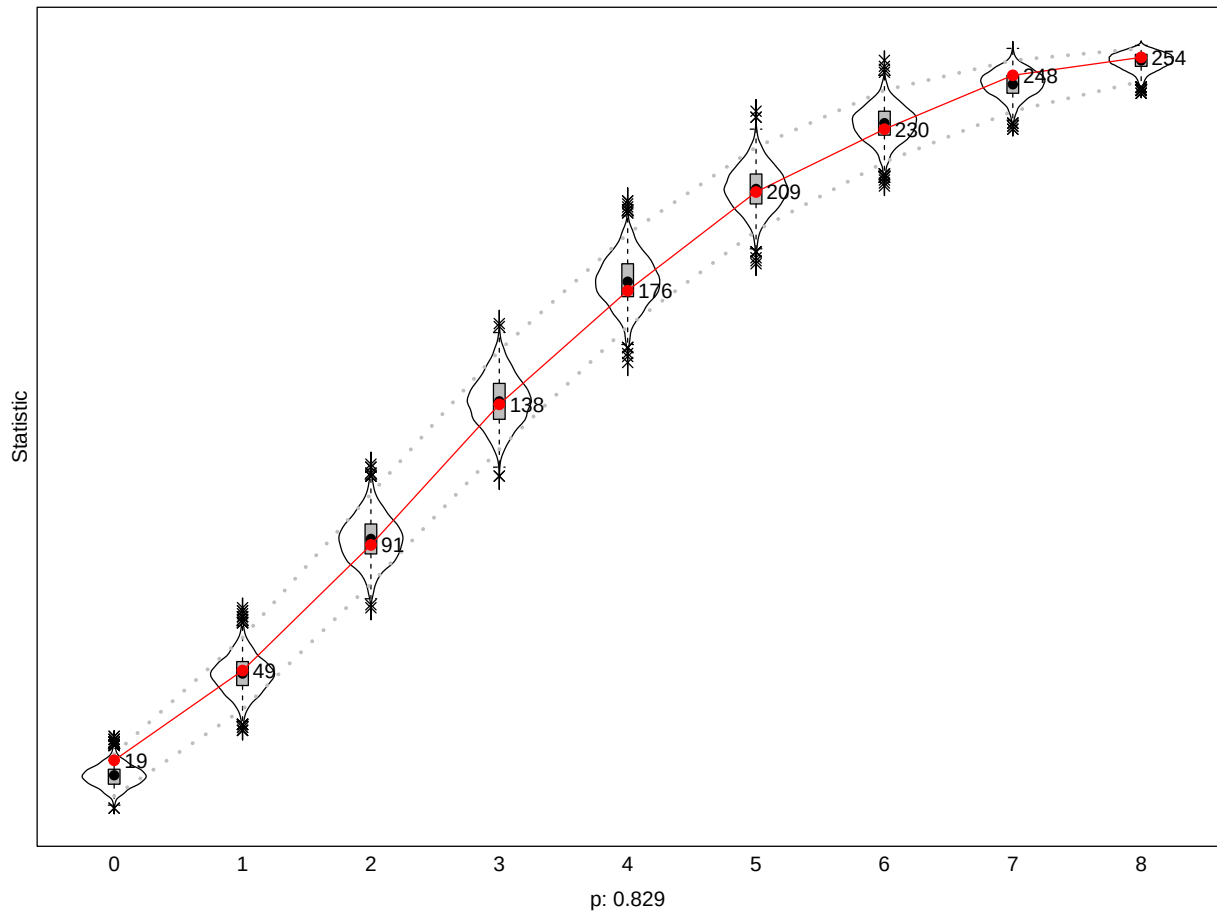
# Geodesic distance
gofEvGD <- test_gof(
  modelEv,
  verbose = FALSE,
  varName = "friendship", GeodesicDistribution
)

stopCluster(cl)

# Plots
plot(gofEvId)

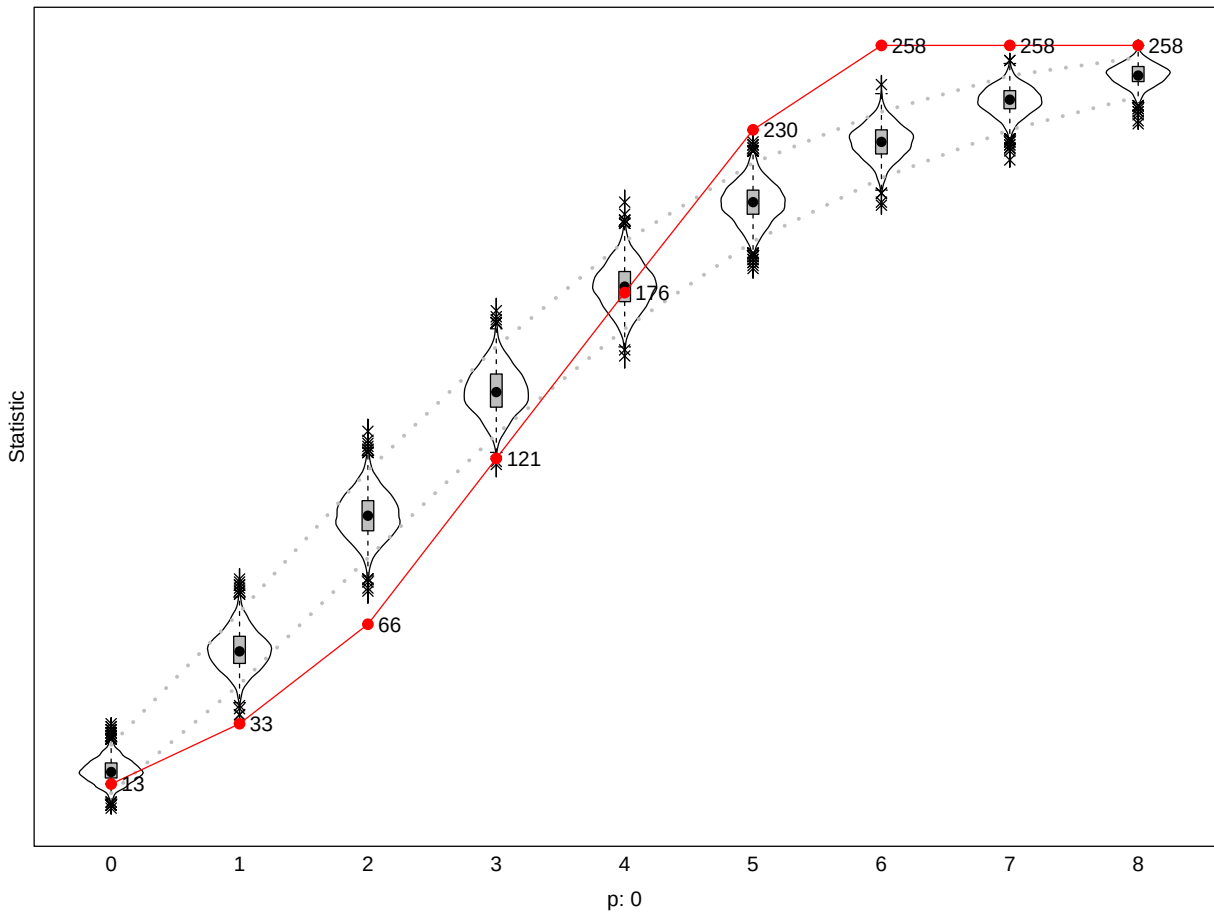
```

Goodness of Fit of IndegreeDistribution



```
plot(gofEvOd)
```

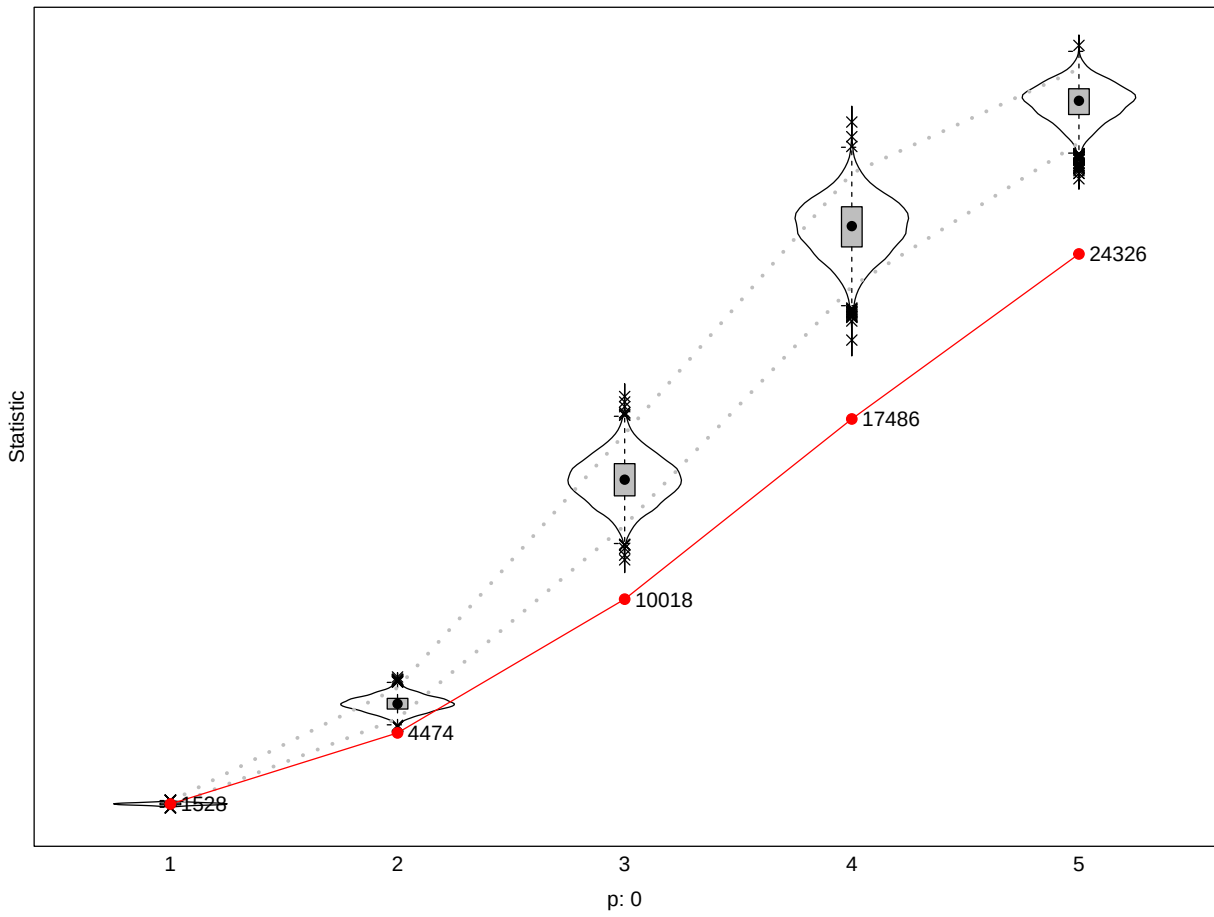
Goodness of Fit of OutdegreeDistribution



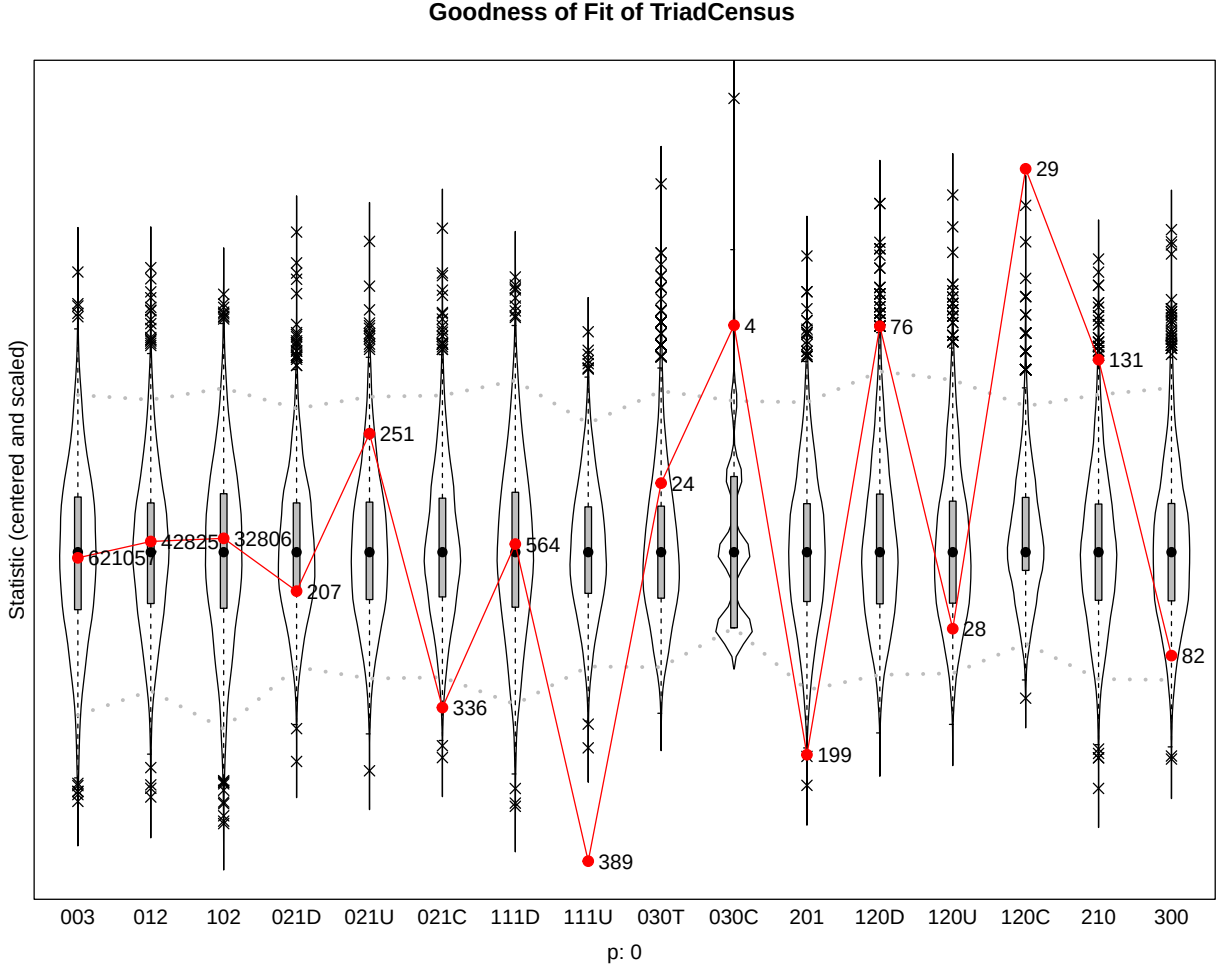
```
plot(gofEvGD)
```

```
## Note: some statistics are not plotted because their variance is 0.  
## This holds for the statistic: Inf.
```

Goodness of Fit of GeodesicDistribution



```
plot(gofEvTC, center = TRUE, scale = TRUE)
```



Comments: Looking at the plot for the goodness of fit of the indegree distribution, we observe that the red line, i.e. the observed network is almost entirely inside the grey distributions, which are the simulated networks, across all indegree levels. This shows that the model fits the indegree distribution very well and successfully captures the distribution of the popularity in the network. Also, the high p-value of 0.829 indicates that there is no significant deviation between the observed distributions and the simulated ones.

In contrast, in the plot for the goodness of fit of the outdegree distribution we see a poor model fit. The red line (observed network) is mostly outside the range of grey distributions (simulated distributions). This suggests that the model doesn't adequately represent the distribution of outgoing ties among students. This is likely due to the lack of an outdegree effect in the model. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the outdegree distribution.

In the plot for the goodness of fit of the geodesic distance distribution we see a poor model fit. The red line (observed network) is mostly outside the range of grey distributions (simulated distributions) and the observed values are consistently lower than the simulated values. More specifically, the observed network seems to have shorter path lengths between actors compared to the simulated one. This suggests that the model doesn't adequately represent the geodesic distance distribution. This is likely due to the lack of a connectivity effect which would capture how students connect across different social groups. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the geodesic distance distribution.

The plot for goodness of fit of the triad census also shows a poor fit, since several observed triad counts are

outside the range of the simulated distributions. The p-value of 0 further supports that the model doesn't accurately represent the triad census distribution and doesn't fully capture all the aspects of the local network structure, even though it includes transitive triplets and three-cycles.

Overall, we see that although the model successfully captures the distribution of popularity, it fails to reproduce the outdegree, the geodesic distance and the triad census distributions. This suggests that additional effects may be required to capture the full structure of the network.

Task 3.1.5

To evaluate the hypotheses i-iii, we examine the estimated coefficients and their significance. The statistical significance can be measured by dividing the estimated coefficient by its standard error.

For Hypothesis i, we were testing whether students tend to be friends with popular pupils. Our results showed that the indegree popularity effect is negative and statistically significant, indicating that students don't tend to form ties with pupils who are more popular. This suggests that students tend to avoid highly popular pupils, or that the formation of ties is not driven by popularity. Therefore, Hypothesis i is not supported by the data.

For Hypothesis ii, we were testing whether students tend to be friends with pupils with similar alcohol consumption levels. To do this, we looked at the alcohol similarity effect. The estimate for this coefficient is positive and statistically significant, indicating that students are more likely to form ties with other students whose alcohol consumption levels are similar to theirs. This shows evidence for homophily based on alcohol consumption and therefore Hypothesis ii is supported by the data,

For Hypothesis iii, we were testing whether student tend to be friends with students who live in the same neighbourhood. For this, we examined the effect of log-distance between student's homes. From our results, the estimated coefficient is negative and statistically significant, indicating that that students are more likely to form ties with pupils whose distance between their homes is shorter. This supports the idea that physical proximity increases the probability of forming a friendship and therefore Hypothesis iii is supported by the data.

Looking at the magnitude of the effects, the alcohol similarity effect (0.638) is notably larger than the distance effect (-0.157), suggesting that behavioral homophily plays a stronger role in friendship formation than geographical proximity.

Overall, the results show that the formation of ties between students is driven by similarity in behaviours such as alcohol consumption as well as geographical proximity, rather than popularity. Students showed a tendency to befriend others who had similar alcohol consumption levels and lived closer to them, while there is no evidence to support the hypothesis that students are more likely to form ties with more popular pupils. In general, this indicates a significance of homophily in the formation of student friendships.

Task 3.2.1

We investigate the co-evolution of friendship and alcohol consumption, by using the model specification developed in (1).

```
# Dependent variables
friendship <- sienaDependent(array(c(f1, f2, f3), dim = c(129, 129, 3)))
alcoholDep <- sienaDependent(alcohol, type = "behavior")

# Covariates
gender <- coCovar(demographic$gender)
age <- coCovar(demographic$age)
distance <- coDyadCovar(logdistance)
```

```

# Siena data object
mydataCoev <- sienaDataCreate(friendship, alcoholDep, gender, age, distance)

# Effects object
myeffCoev <- getEffects(mydataCoev)

## Selection part: friendship evolution -----

# Basic endogenous controls
myeffCoev <- includeEffects(myeffCoev, recip, transTrip, cycle3,
                           name = "friendship")

```

```

##   effectNumber effectName      shortName include fix
## 1 18          reciprocity      recip    TRUE  FALSE
## 2 24          transitive triplets transTrip TRUE  FALSE
## 3 44          3-cycles          cycle3   TRUE  FALSE
##   test  initialValue parm
## 1 FALSE          0    0
## 2 FALSE          0    0
## 3 FALSE          0    0

```

```

# Gender controls
myeffCoev <- includeEffects(myeffCoev, egoX, altX, sameX,
                           interaction1 = "gender",
                           name = "friendship")

```

```

##   effectNumber effectName  shortName include fix  test
## 1 298          gender alter altX      TRUE  FALSE FALSE
## 2 313          gender ego   egoX      TRUE  FALSE FALSE
## 3 369          same gender sameX      TRUE  FALSE FALSE
##   initialValue parm
## 1          0    0
## 2          0    0
## 3          0    0

```

```

# Age controls
myeffCoev <- includeEffects(myeffCoev, egoX, altX, simX,
                           interaction1 = "age",
                           name = "friendship")

```

```

##   effectNumber effectName      shortName include fix  test
## 1 482          age alter      altX      TRUE  FALSE FALSE
## 2 497          age ego        egoX      TRUE  FALSE FALSE
## 3 543          age similarity simX      TRUE  FALSE FALSE
##   initialValue parm
## 1          0    0
## 2          0    0
## 3          0    0

```

```

# Hypothesis i:
myeffCoev <- includeEffects(myeffCoev, inPop,
                           name = "friendship")

##   effectNumber effectName          shortName include
## 1 81           indegree - popularity inPop      TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE           0    0

# Hypothesis ii:
myeffCoev <- includeEffects(myeffCoev, egoX, altX, simX,
                           interaction1 = "alcoholDep",
                           name = "friendship")

##   effectNumber effectName          shortName include
## 1 666           alcoholDep alter      altX      TRUE
## 2 681           alcoholDep ego        egoX      TRUE
## 3 727           alcoholDep similarity simX      TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE           0    0
## 2 FALSE FALSE           0    0
## 3 FALSE FALSE           0    0

# Hypothesis iii:
myeffCoev <- includeEffects(myeffCoev, X,
                           interaction1 = "distance",
                           name = "friendship")

##   effectNumber effectName shortName include fix   test
## 1 265           distance  X           TRUE   FALSE FALSE
##   initialValue parm
## 1           0    0

# Basic behavioural controls
myeffCoev <- includeEffects(myeffCoev, linear, quadratic,
                           name = "alcoholDep")

##   effectNumber effectName          shortName include
## 1 910           alcoholDep linear shape linear      TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE    0.34704    0

# Hypothesis iv:
myeffCoev <- includeEffects(myeffCoev, indeg,
                           name = "alcoholDep",
                           interaction1 = "friendship")

##   effectNumber effectName          shortName include fix
## 1 980           alcoholDep indegree indeg      TRUE   FALSE
##   test  initialValue parm
## 1 FALSE           0    0

```


Hypothesis v:

```
myeffCoev <- includeEffects(myeffCoev, avAlt,
                             name = "alcoholDep",
                             interaction1 = "friendship")
```

```
##   effectNumber effectName          shortName include
## 1 999          alcoholDep average alter avAlt     TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE           0     0
```

myeffCoev

```
##   name          effectName          include
## 1 friendship constant friendship rate (period 1) TRUE
## 2 friendship constant friendship rate (period 2) TRUE
## 3 friendship outdegree (density)      TRUE
## 4 friendship reciprocity              TRUE
## 5 friendship transitive triplets       TRUE
## 6 friendship 3-cycles                  TRUE
## 7 friendship indegree - popularity     TRUE
## 8 friendship distance                  TRUE
## 9 friendship gender alter              TRUE
## 10 friendship gender ego               TRUE
## 11 friendship same gender              TRUE
## 12 friendship age alter                 TRUE
## 13 friendship age ego                   TRUE
## 14 friendship age similarity            TRUE
## 15 friendship alcoholDep alter         TRUE
## 16 friendship alcoholDep ego           TRUE
## 17 friendship alcoholDep similarity     TRUE
## 18 alcoholDep rate alcoholDep (period 1) TRUE
## 19 alcoholDep rate alcoholDep (period 2) TRUE
## 20 alcoholDep alcoholDep linear shape  TRUE
## 21 alcoholDep alcoholDep quadratic shape TRUE
## 22 alcoholDep alcoholDep indegree      TRUE
## 23 alcoholDep alcoholDep average alter  TRUE
##   fix   test  initialValue parm
## 1 FALSE FALSE    7.45424    0
## 2 FALSE FALSE    6.82927    0
## 3 FALSE FALSE   -1.61299    0
## 4 FALSE FALSE    0.00000    0
## 5 FALSE FALSE    0.00000    0
## 6 FALSE FALSE    0.00000    0
## 7 FALSE FALSE    0.00000    0
## 8 FALSE FALSE    0.00000    0
## 9 FALSE FALSE    0.00000    0
## 10 FALSE FALSE   0.00000    0
## 11 FALSE FALSE   0.00000    0
## 12 FALSE FALSE   0.00000    0
## 13 FALSE FALSE   0.00000    0
## 14 FALSE FALSE   0.00000    0
## 15 FALSE FALSE   0.00000    0
## 16 FALSE FALSE   0.00000    0
```

```
## 17 FALSE FALSE      0.00000  0
## 18 FALSE FALSE      0.91000  0
## 19 FALSE FALSE      1.09314  0
## 20 FALSE FALSE      0.34704  0
## 21 FALSE FALSE      0.00000  0
## 22 FALSE FALSE      0.00000  0
## 23 FALSE FALSE      0.00000  0
```

For the behaviour part of the model, we include linear and quadratic shape effects as basic controls for the distribution of alcohol consumption. The linear effect captures the overall trend in alcohol consumption, while the quadratic effect captures whether students tend to avoid extreme values. To test hypothesis iv, we include the indegree effect on alcohol consumption, which captures whether students who receive many friendship nominations tend to increase their alcohol consumption level. To test hypothesis v, we include the average alter effect, which captures whether students tend to adjust their alcohol consumption towards the average level of their friends.

Task 3.2.2

We estimate the model:

```
# Estimation algorithm
myAlgorithm <- set_algorithm_saom(seed = 1908, n3 = 3000, nsub = 4)

# Estimate co-evolution model
modelCoev <- siena(data = mydataCoev,
  effects = myeffCoev,
  returnDeps = TRUE,
  control_algo = myAlgorithm)
```

siena will create/use an output file mydataCoev_out.txt .

```
# Results
summary(modelCoev)
```

```
## Estimates, standard errors and convergence t-ratios
##
##
##
##
## Network Dynamics
## 1. rate constant friendship rate (period 1) 10.8773 ( 1.0002 ) -0.0525
## 2. rate constant friendship rate (period 2) 8.9740 ( 0.8321 ) 0.0456
## 3. eval outdegree (density) -3.1092 ( 0.1322 ) -0.0401
## 4. eval reciprocity 2.1954 ( 0.1139 ) -0.0302
## 5. eval transitive triplets 0.6815 ( 0.0451 ) -0.0746
## 6. eval 3-cycles -0.4604 ( 0.0909 ) -0.0700
## 7. eval indegree - popularity -0.0792 ( 0.0201 ) -0.0406
## 8. eval distance -0.1621 ( 0.0458 ) 0.0060
## 9. eval gender alter -0.1736 ( 0.1081 ) -0.0261
## 10. eval gender ego 0.1444 ( 0.1072 ) -0.0066
## 11. eval same gender 0.8985 ( 0.0982 ) -0.0268
## 12. eval age alter 0.1175 ( 0.1166 ) -0.0318
## 13. eval age ego -0.2221 ( 0.1130 ) 0.0100
```

```

## 14. eval age similarity          0.0477 ( 0.1970 )    0.0356
## 15. eval alcoholDep alter       0.0462 ( 0.0497 )    0.0064
## 16. eval alcoholDep ego        -0.0431 ( 0.0481 )   -0.0024
## 17. eval alcoholDep similarity  1.2344 ( 0.3760 )   -0.0125
##
## Behavior Dynamics
## 18. rate rate alcoholDep (period 1) 1.5555 ( 0.2735 )   -0.0121
## 19. rate rate alcoholDep (period 2) 2.1851 ( 0.3526 )   -0.0224
## 20. eval alcoholDep linear shape    0.1293 ( 0.2574 )   -0.0096
## 21. eval alcoholDep quadratic shape -0.4258 ( 0.1250 )   -0.0334
## 22. eval alcoholDep indegree       0.0741 ( 0.0685 )   -0.0053
## 23. eval alcoholDep average alter   1.0316 ( 0.3671 )   -0.0170
##
## Overall maximum convergence ratio: 0.1554
##
##
## Total of 5815 iteration steps.
##
## Covariance matrix of estimates (correlations below diagonal)
##
##      1.000      -0.058      0.020      -0.001      -0.007      0.009      0.000      0.000
##     -0.069      0.692      0.017      -0.004      -0.003      0.003     -0.001      0.000
##      0.148      0.157      0.017      -0.003      0.000      0.001     -0.002      0.000
##     -0.006     -0.037     -0.190      0.013      0.002     -0.006      0.000      0.000
##     -0.165     -0.079     -0.014      0.380      0.002     -0.003      0.000      0.000
##      0.094      0.046      0.063     -0.599     -0.792      0.008      0.000      0.000
##      0.020     -0.031     -0.638      0.084     -0.247      0.039      0.000      0.000
##      0.035      0.025     -0.034     -0.024     -0.072      0.071      0.135      0.000
##      0.073      0.049      0.072      0.000      0.017     -0.087      0.098      0.000
##     -0.099     -0.020     -0.124     -0.023     -0.026      0.088     -0.018     -0.000
##     -0.111     -0.096     -0.648     -0.141     -0.093      0.110      0.030      0.000
##      0.045     -0.033     -0.072      0.076      0.004     -0.012      0.032      0.000
##     -0.061      0.044      0.024     -0.081      0.034     -0.033     -0.002      0.000
##      0.002     -0.004     -0.013      0.003      0.000     -0.015     -0.034     -0.000
##     -0.010     -0.050     -0.039      0.054     -0.051      0.065     -0.041      0.000
##      0.038      0.023      0.012      0.009      0.112     -0.107      0.019     -0.000
##     -0.184     -0.126     -0.186     -0.050      0.046     -0.050     -0.026     -0.000
##     -0.032     -0.055     -0.020     -0.074     -0.080      0.084      0.010     -0.000
##     -0.045      0.069     -0.015      0.011     -0.055      0.031      0.048      0.000
##      0.010     -0.047     -0.012      0.001      0.039     -0.022      0.000     -0.000
##      0.004     -0.076      0.057      0.057     -0.034      0.036      0.005      0.000
##      0.018      0.083      0.013     -0.006     -0.027      0.011     -0.003      0.000
##      0.026      0.073     -0.025     -0.073      0.013     -0.015     -0.022     -0.000
##
## Derivative matrix of expected statistics X by parameters:
##
##      19.182      0.000      1.345      1.531      26.292      8.610      30.173     -0.000
##       0.000      20.442     -2.206     -0.573       9.709       3.184      -3.480     -0.000
##     142.780     124.776     1173.673      795.515     2962.116      833.695     10026.187   -187.500
##      15.385       5.925      436.761      498.276     1350.191      456.456      3735.563   -70.000
##     249.302     150.778     1845.756     1548.648     9512.848     2819.166     18755.603  -363.400
##      58.421      40.911      800.013      840.753     4322.030     1457.725      8098.324  -178.000
##     794.468     640.985     5647.447     3827.114     16853.871     4693.736     54274.356 -1058.000
##     -35.657     -23.131     -163.693     -121.700     -585.592     -168.844     -1756.888    519.000

```

```

##      -15.169      -9.787      -21.056      -0.559       8.989      15.176      -378.113      -6.2
##      -12.936      -0.754      -17.514      -13.676      -30.696      -8.755      -317.968      -7.4
##      146.112      121.346      1059.112      738.688      2653.057      738.599      9005.882      -173.5
##       4.018       1.196       2.196      -12.060      -4.032      -6.173       6.849      -0.9
##      -8.441      -2.564      -7.382       12.734      16.534      12.870      -60.055      -3.2
##      -0.987      -2.391       7.394       6.277      26.173       7.634      80.480       2.1
##     -17.705      28.219      27.334     -15.617     -37.769     -33.786      61.469     -42.3
##     -38.406      17.644     -62.029     -25.991     -236.970     -50.248     -736.754     -15.6
##      16.876      11.583      77.406      59.007      211.070      64.620      671.474      -3.5
##      -0.405       0.000      -2.802      -0.720       3.386      -1.425     -24.522      1.9
##       0.000      -6.187      -3.305      -1.930      -8.147      -3.242     -32.003     -1.4
##      -2.330       0.351      -7.284      -9.704     -42.616     -13.546     -69.369       6.8
##     -22.422     -18.354     -119.382     -67.378     -178.036     -51.454     -922.701     33.3
##      -8.991       0.689     -16.387     -40.006     -146.668     -62.198     -179.058     18.6
##      -5.547      -9.093     -17.437     -0.788      11.248       6.307     -126.198      7.4
##
## Covariance matrix of X (correlations below diagonal):
##
##      382.363      -0.730      231.257      121.110      915.624      262.584      2447.144      -56.7
##      -0.002      318.007      150.914       55.175      447.096      121.577      1454.441     -35.7
##       0.300       0.214     1558.208     1201.454     5674.755     1666.065     14500.149     -294.7
##       0.176       0.088       0.864     1241.170     5188.900     1661.016     11426.572     -244.4
##       0.254       0.136       0.780       0.799     33971.316     10439.530     61413.363    -1358.5
##       0.231       0.117       0.727       0.812       0.976     3367.933     18115.501     -411.7
##       0.325       0.212       0.955       0.843       0.866       0.811     148000.994    -3126.6
##      -0.125      -0.086      -0.321      -0.298      -0.316      -0.305      -0.349     542.6
##      -0.004      -0.048       0.013       0.044       0.088       0.096       0.001      -0.6
##       0.005      -0.036       0.039       0.042       0.092       0.086       0.031      -0.6
##       0.267       0.210       0.957       0.834       0.722       0.665       0.904     -0.3
##       0.017       0.004       0.013      -0.019       0.012       0.006       0.012       0.6
##      -0.046      -0.028      -0.084      -0.034      -0.058      -0.040      -0.087       0.6
##      -0.011      -0.006       0.061       0.048       0.049       0.044       0.065     -0.6
##      -0.115      -0.021      -0.168      -0.139      -0.140      -0.135      -0.174     -0.6
##      -0.105      -0.046      -0.218      -0.158      -0.168      -0.151      -0.216       0.6
##       0.115       0.030       0.308       0.298       0.269       0.255       0.299     -0.6
##       0.012       0.002       0.042       0.033       0.053       0.044       0.044     -0.6
##       0.002       0.004       0.017       0.004       0.000      -0.004       0.020     -0.6
##       0.000      -0.001       0.087       0.063       0.069       0.058       0.083     -0.6
##      -0.034      -0.063      -0.091      -0.047      -0.028      -0.022      -0.078       0.6
##       0.010       0.007       0.091       0.066       0.076       0.062       0.092     -0.6
##      -0.015      -0.085       0.013       0.035       0.032       0.034       0.008       0.6
##
# Check convergence
modelCoev$tconv.max

##      [,1]
## [1,] 0.155407

```

Comments: The co-evolution model was estimated using a stochastic approximation algorithm with 3000 phase 3 simulations. The overall maximum convergence ratio is 0.1711, which is below 0.25 and therefore indicates satisfactory convergence. Also, good convergence is indicated by the t-ratios being close to zero, suggesting that the estimation procedure reproduced the observed network behavioural statistics well.

Now let's look at the parameters: Overall, the network parameters appear to be quite consistent with the parameters in the friendship-only model. The outdegree parameter estimate is negative (-3.10, with

standard error 0.13), which indicates that the friendship network is sparse. The reciprocity parameter estimate is positive and statistically significant (2.19, with standard error 0.11), indicating that students tend to reciprocate their friendships. The transitive triplets estimate is also positive and significant (0.68, with standard error 0.04), indicating triadic closure, while the three-cycle effect estimate is negative and significant (-0.46, with standard error 0.08) and suggests a more hierarchical than cyclic structure.

The indegree popularity effect estimate is negative and significant (-0.08, with standard error 0.02), indicating that students are less likely to form ties with popular students. The distance effect estimate is also negative and significant (-0.16, with standard error 0.05), suggesting that students are more likely to form ties with others who live close to them. The gender effect is positive and significant (0.90 with standard error 0.10), indicating strong gender homophily. The alcohol similarity estimate is also positive and significant (1.20, with standard error 0.36), showing that students tend to select friends with similar alcohol consumption levels.

Now looking at the behaviour dynamics: The quadratic shape effect is negative and significant (-0.42 with standard error 0.14), suggesting that students are less likely to have extreme alcohol consumption levels, while the linear shape effect (0.16, with standard error 0.24) is not statistically significant, meaning there is no overall directional trend in alcohol consumption. The indegree effect on alcohol consumption is positive but not significant (0.07, with standard error 0.07), indicating that we have no clear evidence that more popular students drink more. In contrast, the average alter effect is positive and significant (1.03, with standard error 0.40), showing that we have evidence that students tend to adjust their alcohol consumption to that of their friends.

Task 3.2.3

We evaluate the goodness of fit of the estimated model using the following auxiliary statistics: indegree distribution, outdegree distribution, triad census, geodesic distance and behavioral distribution.

```
## Goodness of fit

# Geodesic distance function
GeodesicDistribution <- function(
  i, data, sims, period, groupName,
  varName, levls = c(1:5, Inf), cumulative = TRUE
) {
  x <- networkExtraction(i, data, sims, period, groupName, varName)
  require(sna)
  a <- sna::geodist(symmetrize(x))$gdist

  if (cumulative) {
    gdi <- sapply(levls, function(i) sum(a <= i))
  } else {
    gdi <- sapply(levls, function(i) sum(a == i))
  }

  names(gdi) <- as.character(levls)
  gdi
}

cl <- makeCluster(4)
clusterEvalQ(cl, library(RSiena))
```

```
## [[1]]
## [1] "RSiena"      "stats"       "graphics"    "grDevices"
```

```
## [5] "utils"      "datasets"  "methods"   "base"
##
## [[2]]
## [1] "RSiena"     "stats"     "graphics"  "grDevices"
## [5] "utils"      "datasets"  "methods"   "base"
##
## [[3]]
## [1] "RSiena"     "stats"     "graphics"  "grDevices"
## [5] "utils"      "datasets"  "methods"   "base"
##
## [[4]]
## [1] "RSiena"     "stats"     "graphics"  "grDevices"
## [5] "utils"      "datasets"  "methods"   "base"
```

```
clusterEvalQ(cl, library(sna))
```

```
## [[1]]
## [1] "sna"          "network"    "statnet.common"
## [4] "RSiena"       "stats"      "graphics"
## [7] "grDevices"    "utils"      "datasets"
## [10] "methods"     "base"
##
## [[2]]
## [1] "sna"          "network"    "statnet.common"
## [4] "RSiena"       "stats"      "graphics"
## [7] "grDevices"    "utils"      "datasets"
## [10] "methods"     "base"
##
## [[3]]
## [1] "sna"          "network"    "statnet.common"
## [4] "RSiena"       "stats"      "graphics"
## [7] "grDevices"    "utils"      "datasets"
## [10] "methods"     "base"
##
## [[4]]
## [1] "sna"          "network"    "statnet.common"
## [4] "RSiena"       "stats"      "graphics"
## [7] "grDevices"    "utils"      "datasets"
## [10] "methods"     "base"
```

```
clusterExport(cl, "GeodesicDistribution")
```

```
gofId <- test_gof(
  modelCoev,
  auxiliaryFunction = IndegreeDistribution,
  varName = "friendship",
  tested = FALSE,
  cluster = cl
)

gofOd <- test_gof(
  modelCoev,
  auxiliaryFunction = OutdegreeDistribution,
```

```

    varName = "friendship",
    tested = FALSE,
    cluster = cl
)

gofTC <- test_gof(
  modelCoev,
  auxiliaryFunction = TriadCensus,
  varName = "friendship",
  tested = FALSE,
  cluster = cl
)

gofGD <- test_gof(
  modelCoev,
  auxiliaryFunction = GeodesicDistribution,
  varName = "friendship",
  tested = FALSE
)

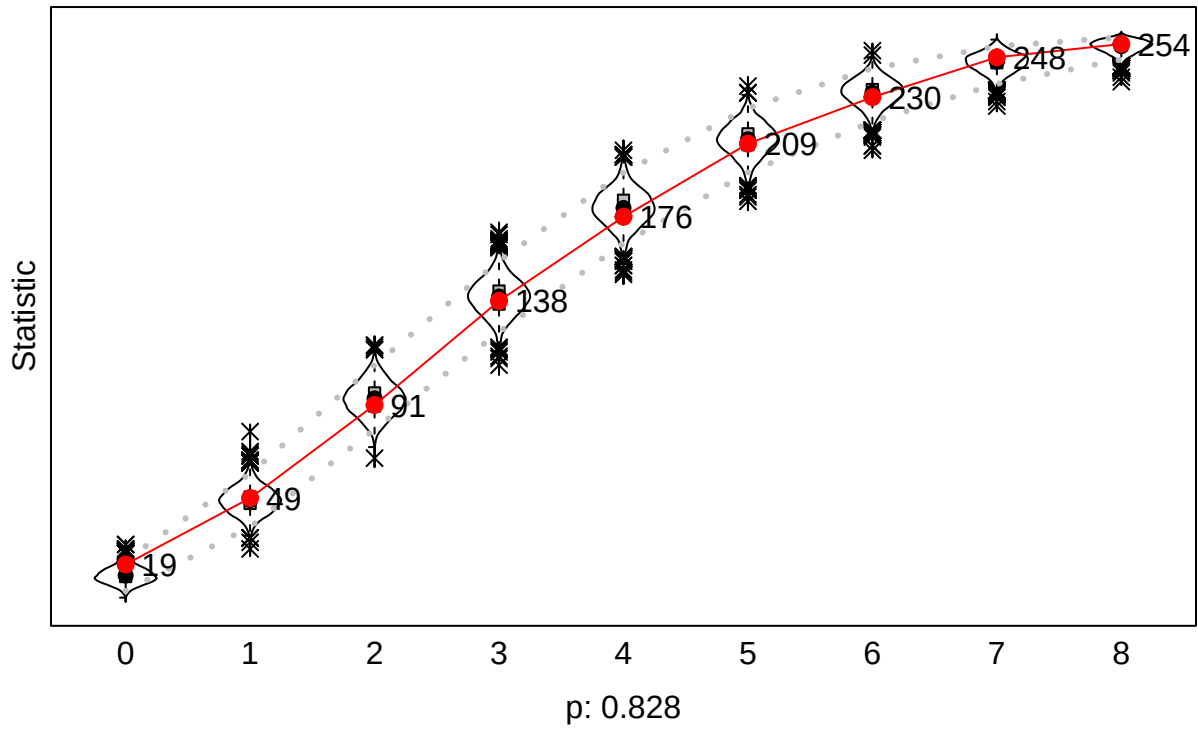
gofBeh <- test_gof(
  modelCoev,
  auxiliaryFunction = BehaviorDistribution,
  varName = "alcoholDep",
  tested = FALSE,
  cluster = cl
)

stopCluster(cl)

# Plots
plot(gofId)

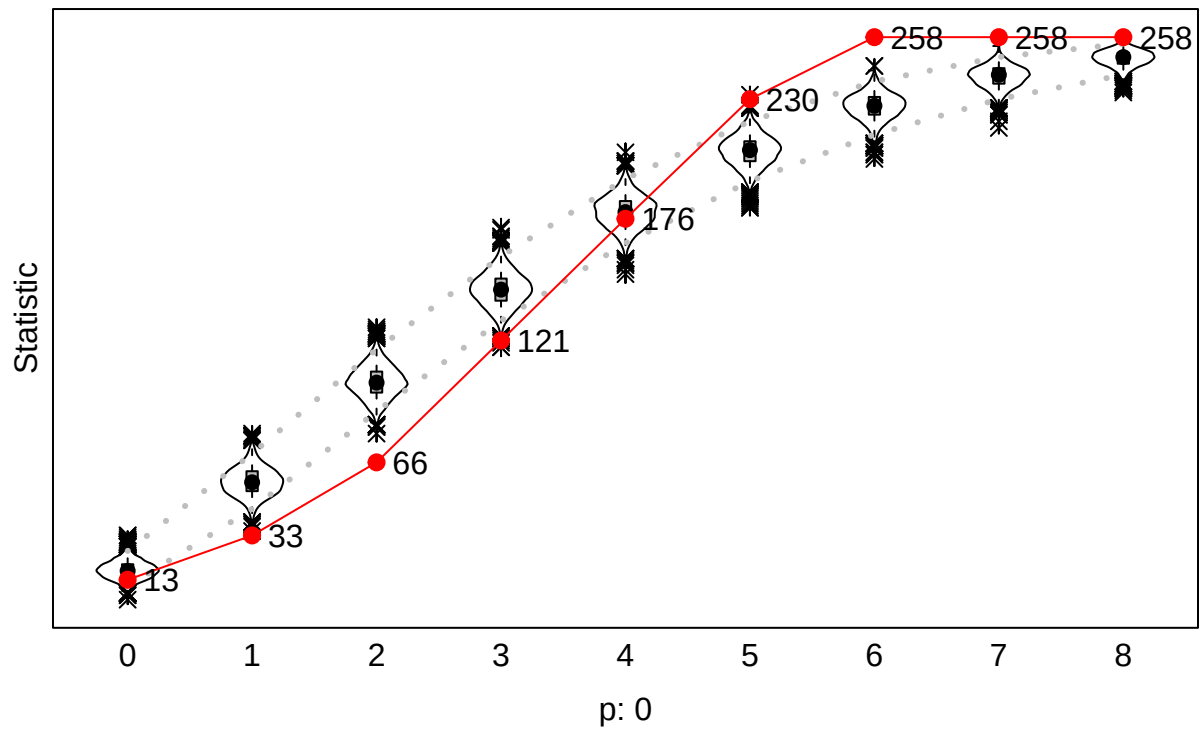
```

Goodness of Fit of IndegreeDistribution



```
plot(gof0d)
```

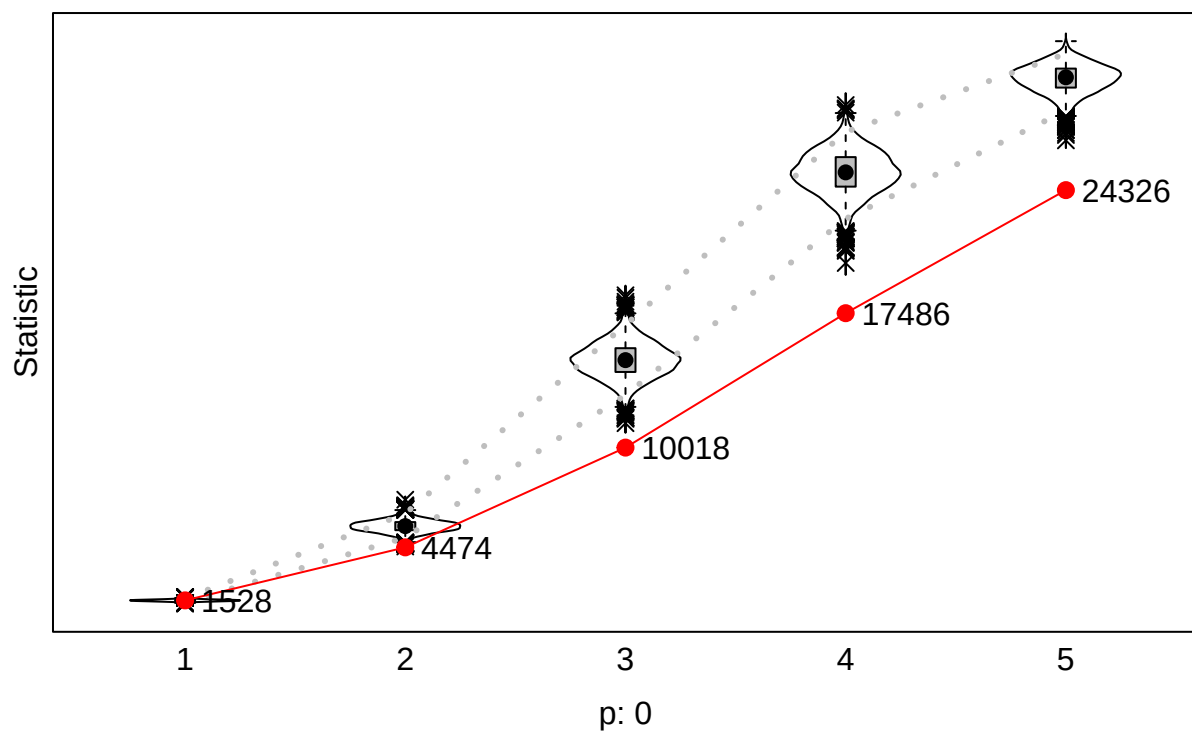

Goodness of Fit of OutdegreeDistribution



```
plot(gofGD)
```

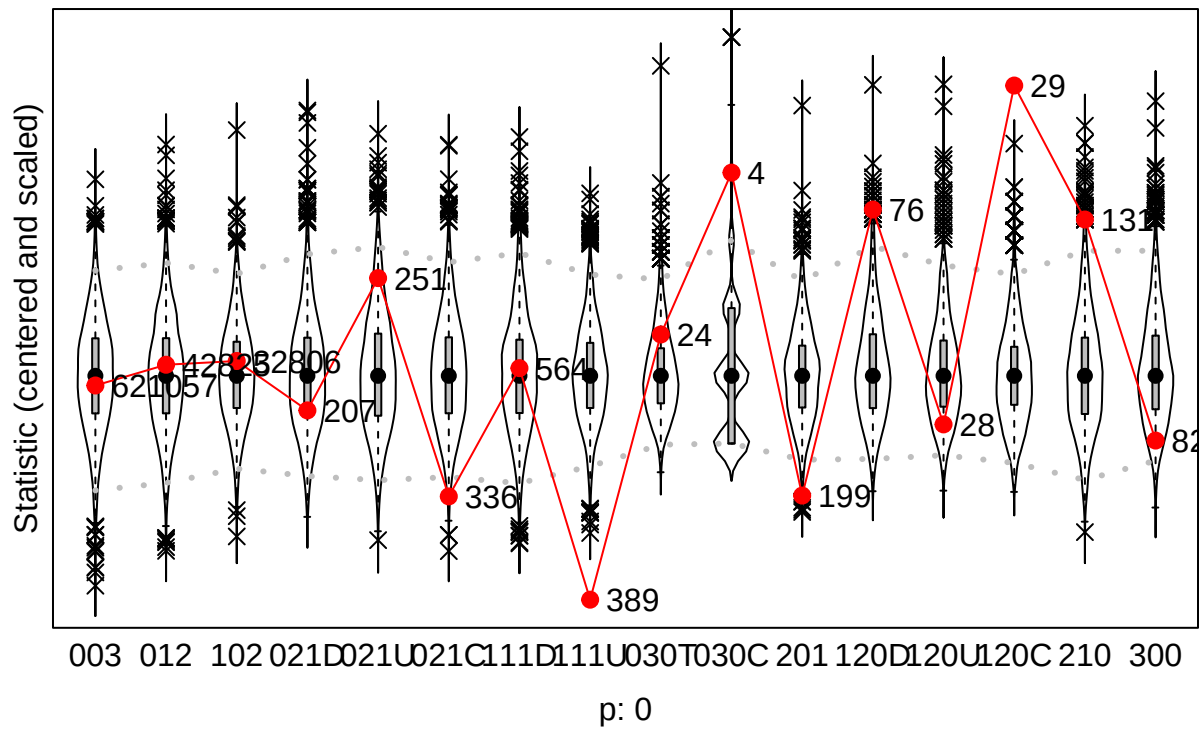
```
## Note: some statistics are not plotted because their variance is 0.  
## This holds for the statistic: Inf.
```

Goodness of Fit of GeodesicDistribution



```
plot(gofTC, center = TRUE, scale = TRUE)
```

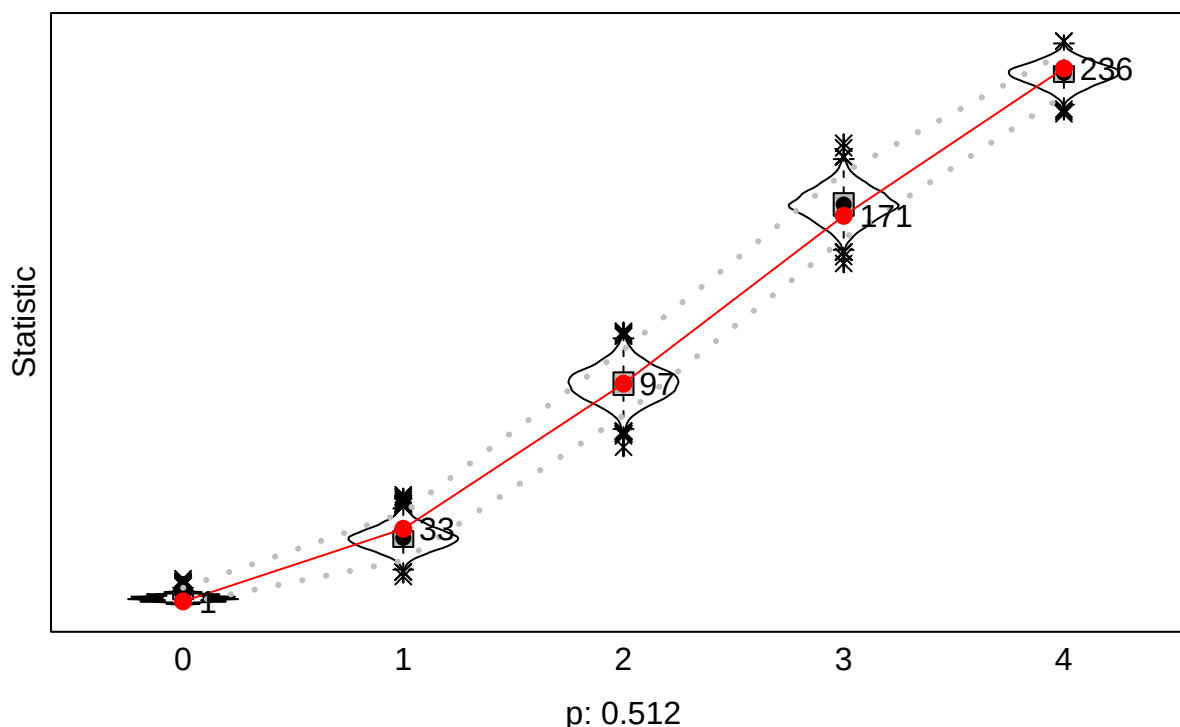
Goodness of Fit of TriadCensus



```
plot(gofBeh)
```

```
## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: 5.
```

Goodness of Fit of BehaviorDistribution



Comments: Looking at the plot for the goodness of fit of the indegree distribution, we observe that the red line, i.e. the observed network is almost entirely inside the grey distributions, which are the simulated distributions, across all indegree levels. This shows that the model fits the indegree distribution very well and successfully captures the distribution of the popularity in the network. Also, the high p-value of 0.84 further confirms that there is no significant deviation between the observed distributions and the simulated ones.

In contrast, in the plot for the goodness of fit of the outdegree distribution we see a poor model fit. The observed network (red line) lies mostly outside the range of simulated distributions (gray bands). This suggests that the model doesn't adequately represent the distribution of outgoing ties among students. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the outdegree distribution.

In the plot for the goodness of fit of the geodesic distance distribution we see a poor model fit. The observed network (red line) is mostly outside the range of simulated distributions (gray bands) and the observed values are consistently lower than the simulated values. More specifically, the observed network seems to have shorter path lengths between students compared to the simulated one. This suggests that the model doesn't adequately represent the geodesic distance distribution. The p-value of 0 further supports these observations and indicates a significant discrepancy between the observed and simulated distributions and therefore we have strong evidence of model misfit for the geodesic distance distribution.

The plot for goodness of fit of the triad census also shows a poor fit, since several observed triad counts (red line) are outside the range of the simulated distributions (gray bands). The p-value of 0 further supports that the model doesn't accurately represent the triad census distribution and doesn't fully capture all the aspects of the local network structure, even though it includes transitive triplets and three-cycles.

Lastly, the plot for the goodness of fit for the behavior distribution shows that the observed values (red line) lie within the range of the simulated distributions (gray bands) across all levels of alcohol consumption.

This shows that the model successfully captures the distribution of alcohol consumption among students in the network. Also, the high p-value of 0.523 further confirms that there is no significant deviation between the observed distributions and the simulated ones. Therefore, the model provides a very good fit for the behavioural component.

Overall, we see that although the model successfully captures the distribution of popularity and behaviour, it fails to reproduce the outdegree, the geodesic distance and the triad census distributions. This suggests that additional effects may be required to capture the full structure of the network. Comparing these results to the goodness of fit of the friendship-only model from 3.1.4, the co-evolution model performs similarly overall. The indegree fit improves slightly, however the outdegree, geodesic distance and triad census remain poorly fitted in both models, indicating that these are persistent structural issues.

Task 3.2.4

To evaluate the hypotheses iv and v, we examine the effects of alcohol consumption in the co-evolution model. For Hypothesis iv, we were testing whether popular students tend to increase maintain their level of alcohol consumption. To do this, we look at the estimate for the indegree effect on alcohol behaviour. From our results, the estimated coefficient was positive (0.07), but it was not statistically significant (the standard error was 0.07). This means that there is no clear evidence to support that popularity increases alcohol consumption and therefore hypothesis iv is not supported by the data.

For Hypothesis v, we were testing whether students tend to adjust their alcohol consumption to that of their friends. To do this, we look at the average alter effect. The estimated coefficient is positive (1.03) and statistically significant (with a standard error of 0.40). This indicates that students tend to adjust their alcohol consumption levels towards the average alcohol consumption level of their friends. This means that we have strong evidence of students influencing each other in alcohol consumption, and therefore hypothesis v is supported by the data.

Comparing the results above with the results from hypotheses i-iii from Task 3.1, we observe that the conclusions are quite similar between the two models. In both the selection-only model and the co-evolution model, the indegree popularity effect is negative and statistically significant, indicating that students do not tend to form friendships with popular pupils. Similarly, the alcohol similarity effect remains positive and significant, suggesting that students tend to form ties with students that have similar alcohol consumption levels to them. The distance effect also remains negative and significant, showing that students tend to form friendships with students that live closer to them.

However, these effects do differ slightly in magnitude between the two models. For example, when we look at the alcohol similarity effect, it is stronger in the co-evolution model. This indicates that when behavioural dynamics are accounted for, similarity in alcohol consumption is more important in forming ties with other students than in the selection-only model. The key difference between the two models is that in the selection-only model, although we could observe a significant effect of similarities in alcohol consumption on the formation of friendships between students, we were not able to distinguish whether the similarity in alcohol consumption reflected a tendency to select similar friends or a tendency for friends to become more similar over time. With the co-evolution model, we could make that distinction and from the results, we observed that students both choose friends with similar alcohol consumption, but also adjust their behaviour in alcohol consumption to match that of their friends.

Overall, the results of the models suggest that homophily and social influence both play important roles in forming friendships, while popularity does not appear to drive either friendship formation or alcohol consumption.

Task 3.2.5

From the estimates in the co-evolution model, we find evidence for both selection and influence processes in the evolution of friendships and alcohol consumption.

Regarding selection processes, we observed that the alcohol similarity effect is positive and statistically significant, indicating that students tend to form ties with pupils that have similar alcohol consumption levels to them. Also, the geographical proximity and gender effects were also shown to be significant. This suggests that homophily plays an important role in the formation of friendships.

Additionally, regarding influence processes, the alter effect on alcohol consumption is positive and statistically significant, indicating that students tend to adjust their alcohol consumption levels to that of their friends. This means that once friendships are formed, students tend to be influenced by their friends and adjust their behaviour to match that of their friends.

As we saw, there was no evidence that popularity affects alcohol consumption, as the indegree effect on alcohol behaviour was not statistically significant. However, the indegree effect in the friendship network is negative and significant, indicating that students are less likely to form ties with already popular peers. This suggests that popularity does not play an important role in behavioural influence or positive selection in this network. Influence operates mainly through students' similarities, rather than their popularity status in the network.

Overall, the results show that we have evidence for both selection and influence processes, as students tended to form friendships with students similar to them, but they also tended to adjust their behaviour to match that of their friends. This shows that it is important that we consider both selection and influence processes simultaneously, because if we do not, then we may fail to reach correct conclusions about the role of similarity and social influence in the network and behavioural dynamics.

Task 3.3

From the previous parts of this task, the goodness of fit results for the co-evolution model showed that the model reproduced well the indegree and alcohol behaviour distributions. However, the model fit for the outdegree distribution, the geodesic distances and local structure configurations such as triad census, was poor. Therefore, the model could be improved by adding new effects so that geodesic distances and degree distributions might be better represented.

The poor model fit for the outdegree distribution suggests that some students are much more socially active than others and the current model doesn't account for this variation. Therefore, to improve the fit of the degree distribution, and specifically the outdegree distribution, we could include an outdegree activity effect. Some students are naturally more active in seeking friendships than others, and the current model does not account for this variation.

Also, we could include effects such as outdegree popularity and indegree activity, which would help the model capture the dependence between students' activity and popularity better. These effects could help capture the differences between the number of friendship nominations that students send, but also help represent how popularity influences the likelihood of receiving ties, which may not be fully captured just by the indegree popularity effect.

The poor model fit of the geodesic distances distribution suggests that the observed network is more tightly connected than the simulated ones. In order to improve the representation of geodesic distances, we could add effects that capture broader network connectivity. This could be covariates such as students sharing classes or extracurricular activities, since students that interact more frequently within shared contexts are likely to form ties and reduce the distance between them. Since the model does not include those covariates it underestimates the level of connectivity observed in the real network.

Task 3.4

To further investigate the dynamics of friendship and alcohol consumption using SAOMs we could look at the following 2 hypotheses:

Hypothesis 1: Higher alcohol consumption levels increase the likelihood of receiving friendship nominations. (Note that this hypothesis is distinct from hypothesis iv, which tests the opposite causal direction - whether popularity drives alcohol consumption). This hypothesis is based on the idea that engaging in certain behaviours such as alcohol consumption could increase social attractiveness and help in being included in social groups. To test this hypothesis, the alcohol consumption effect should be included as an alter effect in the friendship evaluation function in the following way:

```
includeEffects(myeff, altX, interaction1 = "alcoholDep", name = "friendship")
```

Then after estimating the model, a positive and statistically significant coefficient would indicate that students with higher alcohol consumption are more likely to be more popular. This effect could improve the model fit by capturing behavioural mechanisms influencing popularity and hence help the model better represent the indegree distribution.

Hypothesis 2: Students that attend the same classes are more likely to form friendships. This hypothesis is based on the idea that students who interact more often and happen to be in the same environments are more likely to communicate, develop social relationships and form friendship ties among them. For this hypothesis, we would have to collect data on whether two students attend a class together or not and construct a binary dyadic covariate which would indicate whether two students share at least one class (1=share at least one class, 0=otherwise). This could then be included in the friendship evaluation function as a dyadic covariate effect in the following way:

```
includeEffects(myeff, X, interaction1 = "sameClass", name = "friendship")
```

Then after estimating the model, a positive and statistically significant coefficient would indicate that sharing classes increases the likelihood of students forming a friendship tie. This effect could improve the model fit by capturing shared social contexts and hence improve the representation of geodesic distances.